AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 29 · STUDENT LESSON

Uncertainty and failure analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Read the score. Check the label. Count every case.

Distinguish confidence from correctness and report review coverage alongside accepted accuracy.

Keep one habit: Six-case ledger → confusion counts → coverage → separate picture → failure investigation → executable audit → independent outputs.

MAIN LESSON · SECTION 1

A confident prediction can still be wrong

> Separate the selected class, its score and checked correctness.

Learning goals

I can distinguish a predicted class, **confidence score** and observed correctness.

I can calculate accuracy and **coverage** for a review policy.

I can investigate failure patterns without claiming an untested cause.

Retrieve evaluation evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Last week a model was correct on two constructed validation examples. Why did that not prove it would work reliably on new examples?

> **My starting explanation**

End goal

By the end, I can explain uncertainty and failure analysis using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Distinguish a predicted class, **confidence score** and observed correctness

Calculate accuracy and **coverage** for a review policy

Investigate failure patterns without claiming an untested cause

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Read a score before using it to make a decision

> The six outputs are supplied classroom data, with reference labels for checking.

Class definition and selected-class confidence

A model can be confidently wrong. Use these supplied **output**s for a fictional image classifier: class 1 means a triangle is present; class 0 means it is absent.

The values p1 are probability estimates for class 1, invented for this exercise rather than produced by a trained model. Predict 1 when p1 >= 0.5 and 0 otherwise. The selected-class confidence is max(p1, 1 - p1).

Complete original scores and reference labels

Case and image	p1	Reference label
A clear	0.95	1
B blurred	0.90	0
C clear	0.80	1
D blurred	0.60	0
E clear	0.45	0
F blurred	0.10	1

Contrast A with confidently wrong F

A predicts 1 with confidence 0.95 and is correct. F predicts 0 with confidence 0.90 and is wrong: p1 = 0.10 is not low confidence in the selected class.

Reference labels are supplied for this exercise; real investigations must check their quality too.

Calibration needs many comparable cases

A score alone cannot establish reliability. **Calibration** asks whether, across many comparable cases, **prediction**s assigned a probability such as 0.90 are correct about 90 percent of the time.

One correct or incorrect case cannot answer that question.

Record given and missing evidence

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Trace every prediction, error and acceptance decision

> Keep the class threshold 0.50 separate from the review cutoff 0.80.

Overall accuracy and confusion counts

The **prediction**s for A through F are 1, 1, 1, 1, 0, 0. A, C and E are correct, so overall accuracy is 3 / 6 = 50 percent.

There are two false positives (B and D), one false negative (F), two true positives and one true negative. Different errors can matter differently; a single accuracy number hides that distinction.

All six decisions in one ledger

Case · p1 · Prediction · Selected-class confidence · Correct? · At cutoff 0.80

> **Case:** A
>
> **p1:** 0.95
>
> **Prediction:** 1
>
> **Selected-class confidence:** 0.95
>
> **Correct?:** Yes
>
> **At cutoff 0.80:** Accept

Case: B

p1: 0.9

Prediction: 1

Selected-class confidence: 0.9

Correct?: No: false positive

At cutoff 0.80: Accept

Case: C

p1: 0.8

Prediction: 1

Selected-class confidence: 0.8

Correct?: Yes

At cutoff 0.80: Accept at equality

Case: D

p1: 0.6

Prediction: 1

Selected-class confidence: 0.6

Correct?: No: false positive

At cutoff 0.80: Defer

Case: E

p1: 0.45

Prediction: 0

Selected-class confidence: 0.55

Correct?: Yes

At cutoff 0.80: Defer

Case: F

> **p1:** 0.1
>
> **Prediction:** 0
>
> **Selected-class confidence:** 0.9
>
> **Correct?:** No: false negative
>
> **At cutoff 0.80:** Accept

Runnable original report

```
rows = [(0.95, 1), (0.90, 0), (0.80, 1),
        (0.60, 0), (0.45, 0), (0.10, 1)]

def report(rows, cutoff):
    if not rows or not 0.5 <= cutoff <= 1:
        raise ValueError("Need rows and cutoff in [0.5, 1]")
    correct = accepted = accepted_correct = 0
    for p1, label in rows:
        if not 0 <= p1 <= 1 or label not in (0, 1):
            raise ValueError("Invalid score or label")
        pred = int(p1 >= 0.5)
        ok = int(pred == label)
        correct += ok
        if max(p1, 1 - p1) >= cutoff:
            accepted += 1
            accepted_correct += ok
    return (correct / len(rows), accepted / len(rows),
            accepted_correct / accepted if accepted else None)

print(report(rows, 0.8))  # (0.5, 4/6, 0.5) approximately
```

Denominators, boundaries and undefined accuracy

The function returns overall accuracy, **coverage** and accuracy among accepted cases. It assumes numeric scores and binary labels. None means no cases were accepted, so accepted-case accuracy is undefined, not zero or perfect.

Keep counts with proportions and declare the comparison at the boundary.

Explain the pictured rule

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Raise a cutoff and count what remains

> Predict the accepted set before calculating proportions.

Current task and response

- For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the changed picture

- Change: Raise the acceptance cutoff to 0.85.

▼ **Check after predicting**

Prediction: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Report accepted accuracy together with coverage and workload

> Deferring a case does not supply a corrected outcome.

Main six-case acceptance and missing reviewer outcomes

Confidence does not remove every error

CASES	AT CUTOFF 0.80	OBSERVED RESULT
A, B, C, F	Accept automatically	2 correct, 2 wrong
D, E	Defer for review	1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Deferred cases are not automatically repaired. A review workflow needs an assigned reviewer, suitable **evidence**, a decision record and an outcome check.

Report review workload and the final system's results separately; we have no reviewer outcomes here. Do not erase deferred cases from the report or count them as correct.

Try main cutoff 0.95

> **Raise the cutoff to 0.95 for the same six cases. Which cases are accepted, what are coverage and accepted-case accuracy, and why does that result not guarantee perfect future predictions?**

Separate three-case picture and complete givens

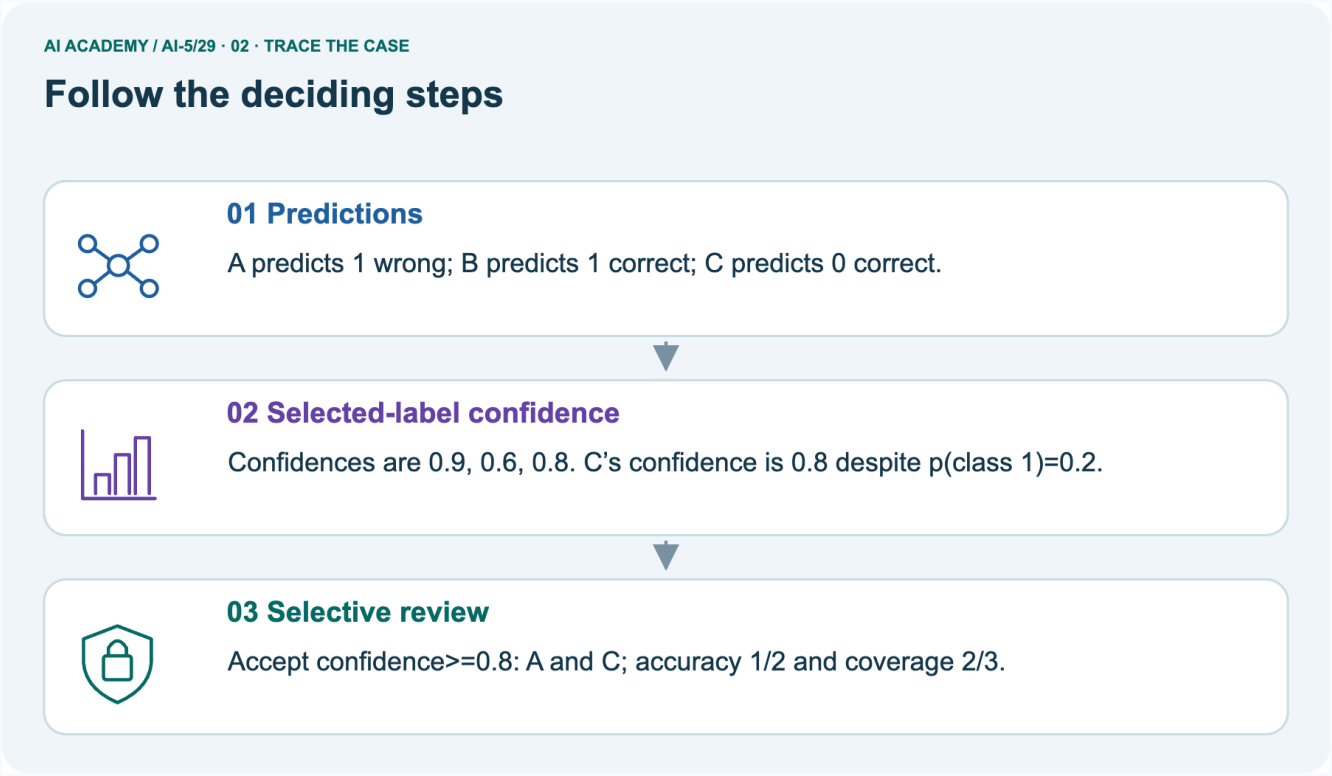
Keep positive-class probability separate from confidence in the chosen label.

Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main six-case dataset and the three-case picture reuse case letters but have different supplied scores and labels. In the picture only, A is wrong at confidence 0.90, B is correct at 0.60 and C is correct at 0.80. Cutoff 0.80 accepts A and C: one correct of two accepted, coverage two of three. Raising this pictured cutoff to 0.85 accepts only the wrong A. Never mix these counts with the main six-case ledger.

Trace confidence and acceptance



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predictions	A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.
2. Selected-label confidence	Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.
3. Selective review	Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

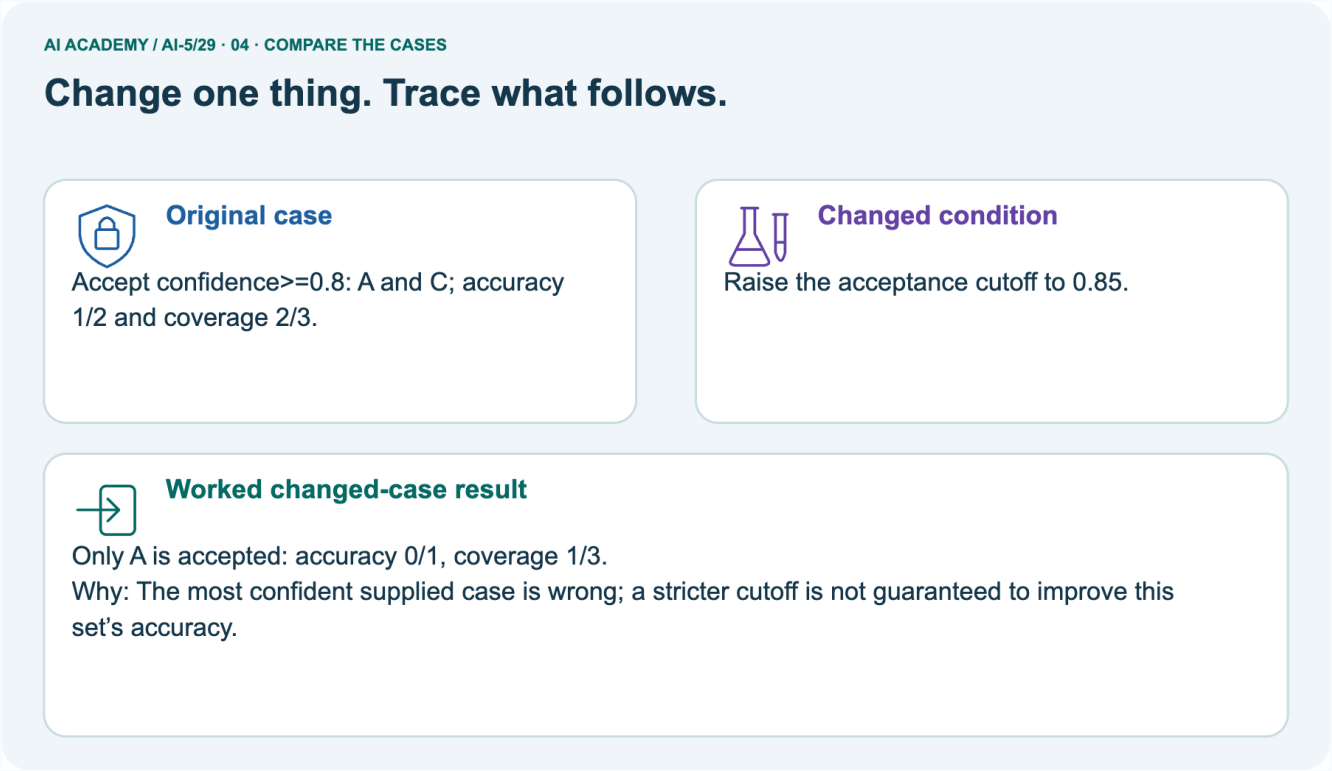
Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Predict the pictured cutoff change

- Change one thing: Raise the acceptance cutoff to 0.85. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain C and retain the limits

- Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

- Read the labelled picture: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

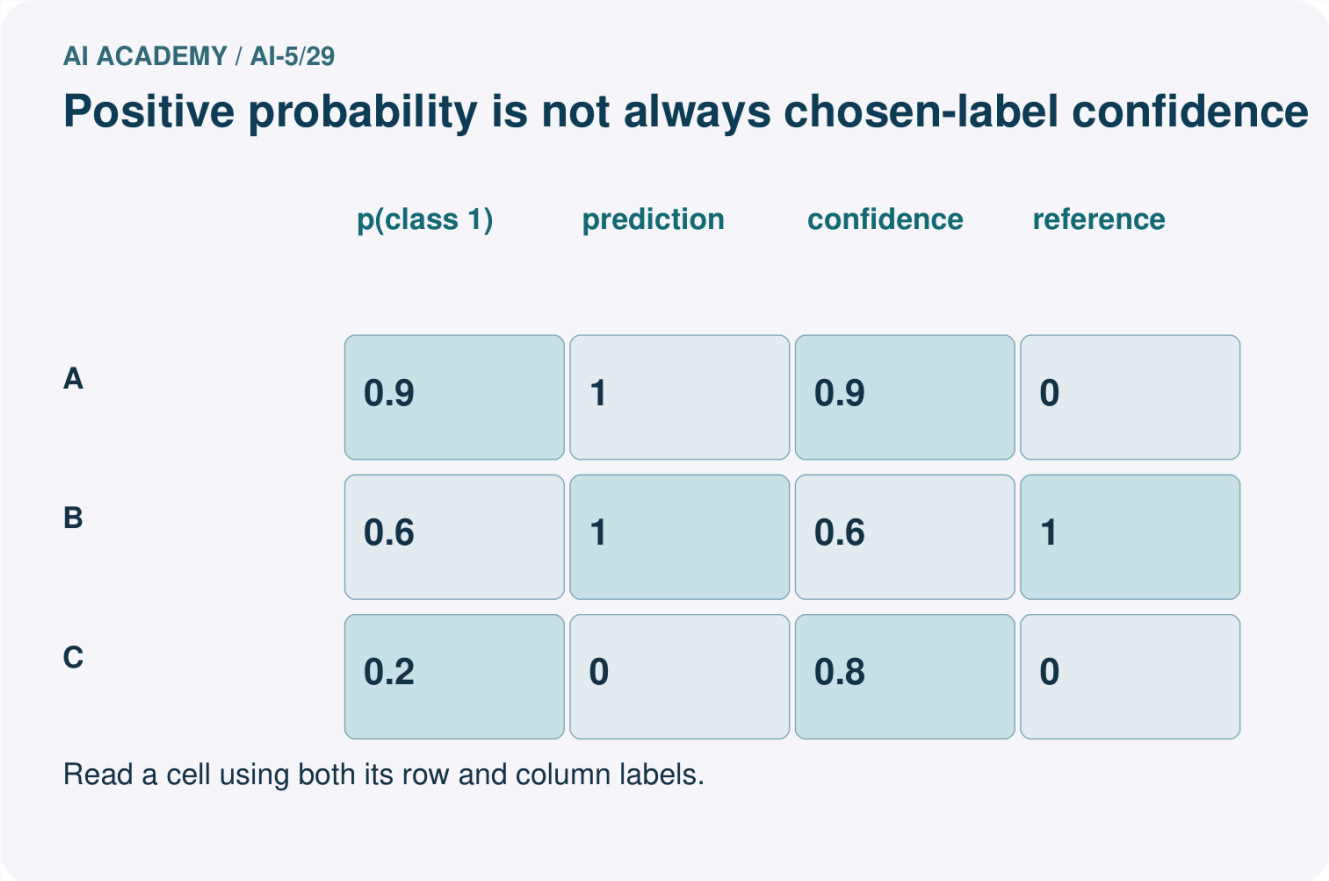
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Uncertainty and failure analysis

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/29

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.



3. Selective review

Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Investigate a failure pattern without inventing its cause

> Use controlled new comparisons and source-group separation.

Observed pattern, proposed investigation and uncertainty

All three clear images are correct; all three blurred images are wrong in this constructed set.

Blur is a candidate explanation, not an established cause: other differences, reference-label errors or pre**process**ing bugs could be involved.

Record case ID, **input** conditions, model and pre**process**ing version, score, **prediction** and checked reference label.

First inspect B, D and F and check **input** order, resizing and label quality. Then plan new examples that vary blur while keeping the underlying image and label fixed.

Keep related versions of the same source image in one data split; compare both versions within the reserved group. Do not repair cases and report scores on those same cases as independent **evidence**.

Uncertainty can arise from ambiguous **input**s, limited **training evidence** or unfamiliar conditions. A single **confidence score** does not identify the cause and may stay high on unfamiliar **input**s.

Collect **evidence** before selecting a remedy.

Evidence and next-check ledger

Observation or proposed step	What the evidence supports
Clear: 3 correct / 3; blurred: 0 / 3	A pattern in these six supplied cases; not a causal conclusion
Check B, D and F	Inspect references, preprocessing, input order and model version
Paired new clear/blurred versions	Keep source image and reference fixed; reserve source groups together

Observation or proposed step	What the evidence supports
Clear original in training, its blurred copy in final evaluation	Related-source overlap; keep the whole source group in one split
Deferred D/E	Two review tasks; no reviewer decisions or final system accuracy supplied
No accepted cases at cutoff 0.99	Coverage 0/6; accepted accuracy undefined, not 0% or 100%

Check accepted cases and missing outcomes

- At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?

  What does the report function return for accepted-case accuracy at cutoff 0.99?

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Practise reading scores and keeping denominators

> A low p1 can mean high confidence in predicted class 0.

Current task and changed cutoff

- For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

> **Raise the cutoff to 0.95 for the same six cases. Which cases are accepted, what are coverage and accepted-case accuracy, and why does that result not guarantee perfect future predictions?**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Uncertainty and failure analysis”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p1 always means low confidence”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Separate picture response

- What is C’s positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve independent K/L/M/N

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Count both types of error

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete decision and workload audit

> Keep a prediction, an actual run or paper trace, and the limits in one record.

Current confusion-count investigation

For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Complete executable multi-cutoff audit

```
"""Invented development scores; no model or human-review run is claimed."""
from math import isfinite
rows = [('A', 'clear', .95, 1), ('B', 'blurred', .90, 0),
        ('C', 'clear', .80, 1), ('D', 'blurred', .60, 0),
        ('E', 'clear', .45, 0), ('F', 'blurred', .10, 1)]
def report(rows, cutoff):
    if not rows or not isfinite(cutoff) or not .5 <= cutoff <= 1:
        raise ValueError('Need rows and finite cutoff in [0.5,1]')
    ledger = []
    for name, condition, p1, label in rows:
        if not isfinite(p1) or not 0 <= p1 <= 1 or label not in (0,1):
            raise ValueError('Invalid score or binary reference')
        pred = int(p1 >= .5)
        confidence = max(p1, 1-p1)
        ledger.append((name, pred, confidence, pred == label,
                       confidence >= cutoff))
    accepted = [r for r in ledger if r[4]]
    return {'ledger': ledger, 'total': len(rows),
            'correct': sum(r[3] for r in ledger),
            'accepted': [r[0] for r in accepted],
            'deferred': [r[0] for r in ledger if not r[4]],
            'coverage': len(accepted)/len(rows),
            'accepted_accuracy': (sum(r[3] for r in accepted)/len(accepted)
                                  if accepted else None)}
for cutoff in [.5, .8, .95, .99]:
    print(cutoff, report(rows, cutoff))
r = report(rows, .8)
assert r['accepted'] == ['A','B','C','F']
```

```
assert r['deferred'] == ['D','E']
assert r['correct'] == 3 and r['accepted_accuracy'] == .5
assert report(rows,.99)['accepted_accuracy'] is None
confusion = {'TP':0, 'TN':0, 'FP':0, 'FN':0}
for _, _, p1, label in rows:
    pred = int(p1 >= .5)
    key = ('TP' if label else 'FP') if pred else ('FN' if label else 'TN')
    confusion[key] += 1
assert confusion == {'TP':2,'TN':1,'FP':2,'FN':1}
print('Confusion counts:', confusion)
for condition in ['clear','blurred']:
    group = [r for r in rows if r[1] == condition]
    print(condition, report(group,.8)['correct'], '/',len(group))
# Separate pictured scores: stricter cutoff keeps the confidently wrong case.
picture = [('A','toy',.9,0),('B','toy',.6,1),('C','toy',.2,0)]
assert report(picture,.8)['accepted_accuracy']==.5
assert report(picture,.85)['accepted_accuracy']==0
print('Picture cutoff .85:',report(picture,.85))
# No human review outcomes or controlled blur comparisons are supplied.
```

Expected records and unavailable outcomes

1. Predict all six decisions before running or tracing. Compare cutoffs 0.50, 0.80, 0.95 and 0.99; save accepted and deferred IDs, counts and both denominators. The class rule stays p1 ≥ 0.50 throughout. At no accepted cases, report accepted accuracy as unavailable. The code validates finite scores and cutoffs, prints confusion counts and subgroup counts, and separately checks the three-case picture.

   It cannot estimate reviewer success, prove blur causes errors, establish calibration or provide independent final evaluation: none of that evidence is supplied.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Reference: earlier report function

```
rows = [(0.95, 1), (0.90, 0), (0.80, 1),
        (0.60, 0), (0.45, 0), (0.10, 1)]

def report(rows, cutoff):
    if not rows or not 0.5 <= cutoff <= 1:
        raise ValueError("Need rows and cutoff in [0.5, 1]")
    correct = accepted = accepted_correct = 0
    for p1, label in rows:
        if not 0 <= p1 <= 1 or label not in (0, 1):
            raise ValueError("Invalid score or label")
        pred = int(p1 >= 0.5)
        ok = int(pred == label)
        correct += ok
        if max(p1, 1 - p1) >= cutoff:
            accepted += 1
            accepted_correct += ok
    return (correct / len(rows), accepted / len(rows),
            accepted_correct / accepted if accepted else None)

print(report(rows, 0.8))  # (0.5, 4/6, 0.5) approximately
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: analyse a fresh four-case score table

> Use the new references; do not assume a stricter cutoff improves accuracy.

Complete independent K/L/M/N inputs

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Analyse new outputs independently

Attempt this without hints

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the failures as well as the successes

> Record dataset, model and preprocessing versions with the review policy.

Project artifact and evidence

Add a labelled uncertainty and failure analysis **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, **confidence score** and observed correctness.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with an honest policy report

> Include coverage, accepted errors, deferred workload and missing outcomes.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Distinguish a predicted class, **confidence score** and observed correctness

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Confidence score	The model’s score for its selected class, which may not match observed reliability
Calibration	Agreement between predicted probabilities and observed frequencies across many comparable cases
Abstention	Withholding an automatic prediction and following a declared alternative process
Coverage	The fraction of all cases on which a system makes an automatic prediction

Use two terms to explain a step in this week’s example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

- At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?

  What does the report function return for accepted-case accuracy at cutoff 0.99?

- All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

  Identify the unsupported claim and the split problem; propose a better investigation.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the fresh independent K/L/M/N outputs for the first individual attempt.

Current explanation and response

- Choose review thresholds and model changes using **training** and validation **evidence**, before final evaluation. Report total cases, overall errors, accepted errors, **coverage**, deferred workload and subgroup results.

  For tiny groups, give counts and avoid strong claims about future rates. Higher cutoffs do not guarantee higher observed accuracy or safe decisions.

- Keep a baseline and record unsuccessful changes too. Next week examines dataset shift: performance can change when the **input**s or relationships encountered in use differ from those seen during development.

- All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

  Identify the unsupported claim and the split problem; propose a better investigation.

> **A teammate reports “100 percent accurate” after accepting only A at cutoff 0.95. Rewrite the claim honestly and name the evidence still needed before trusting this policy.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Confidence does not remove every error

CASES	AT CUTOFF 0.80	OBSERVED RESULT
A, B, C, F	Accept automatically	2 correct, 2 wrong
D, E	Defer for review	1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**
>
> _____
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Keep positive-class probability separate from confidence in the chosen label.

By the end, I can	How I will show it
Distinguish a predicted class, confidence score and observed correctness	Explain the deciding step, show a trace or test, and state one limit.
Calculate accuracy and coverage for a review policy	Explain the deciding step, show a trace or test, and state one limit.
Investigate failure patterns without claiming an untested cause	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Keep positive-class probability separate from confidence in the chosen label.
- Which labelled step matches this detail? Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.
- 1. Selected-label confidence
- 2. Selective review
- 3. **Prediction**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Selective review Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Changed-case question and optional hint

- Changed-case practice: Raise the acceptance cutoff to 0.85. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Only A is accepted: accuracy 0/1, **coverage** 1/3.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Only A is accepted: accuracy 0/1, **coverage** 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Repair and retain the record

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference · Six-picture concept journey</summary>

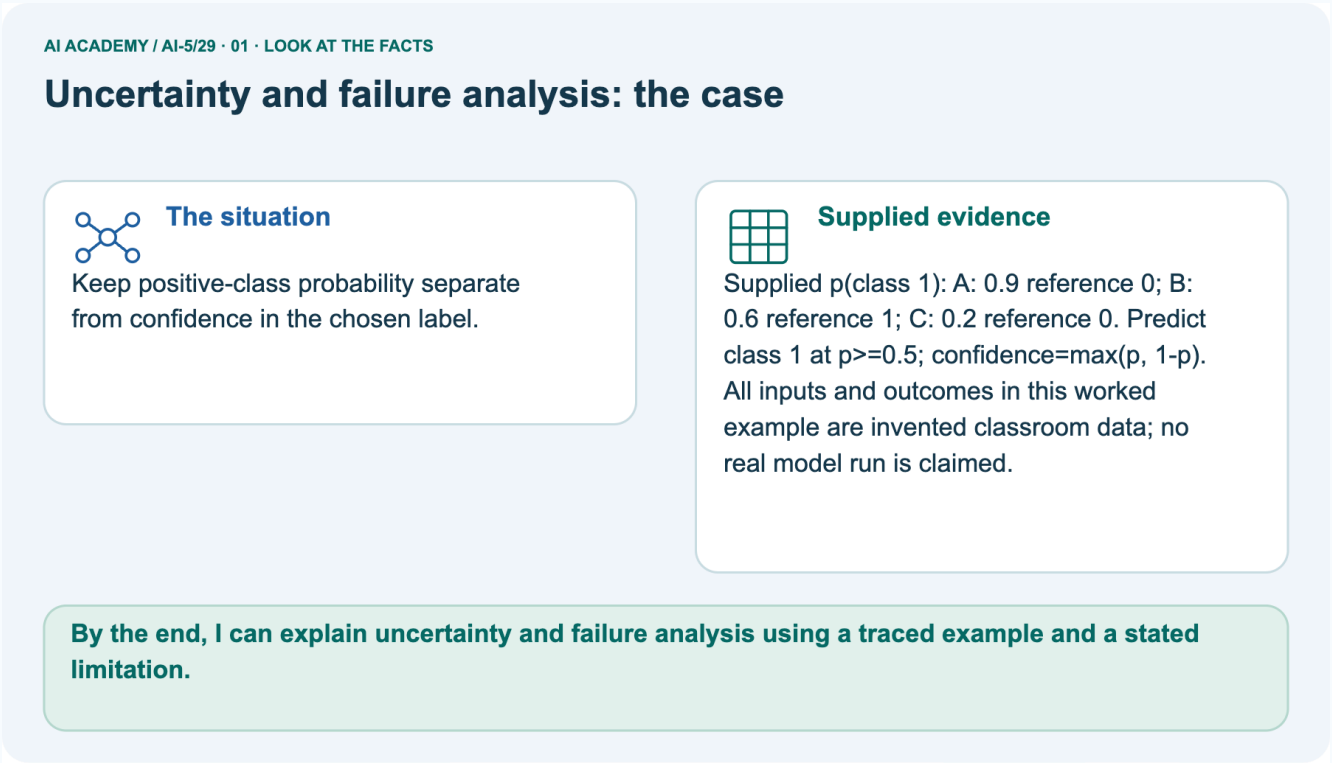
</details>

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Uncertainty and failure analysis: the case



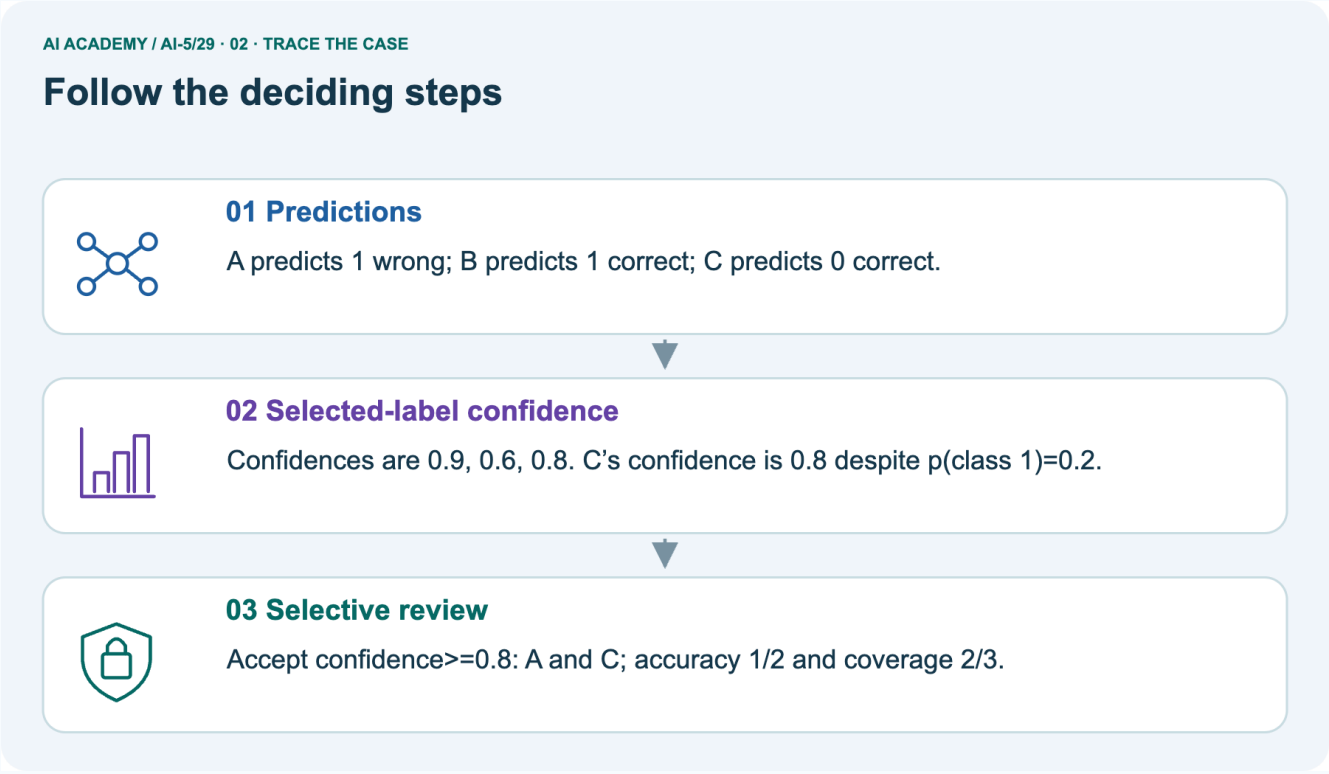
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep positive-class probability separate from confidence in the chosen label. Supplied facts: Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



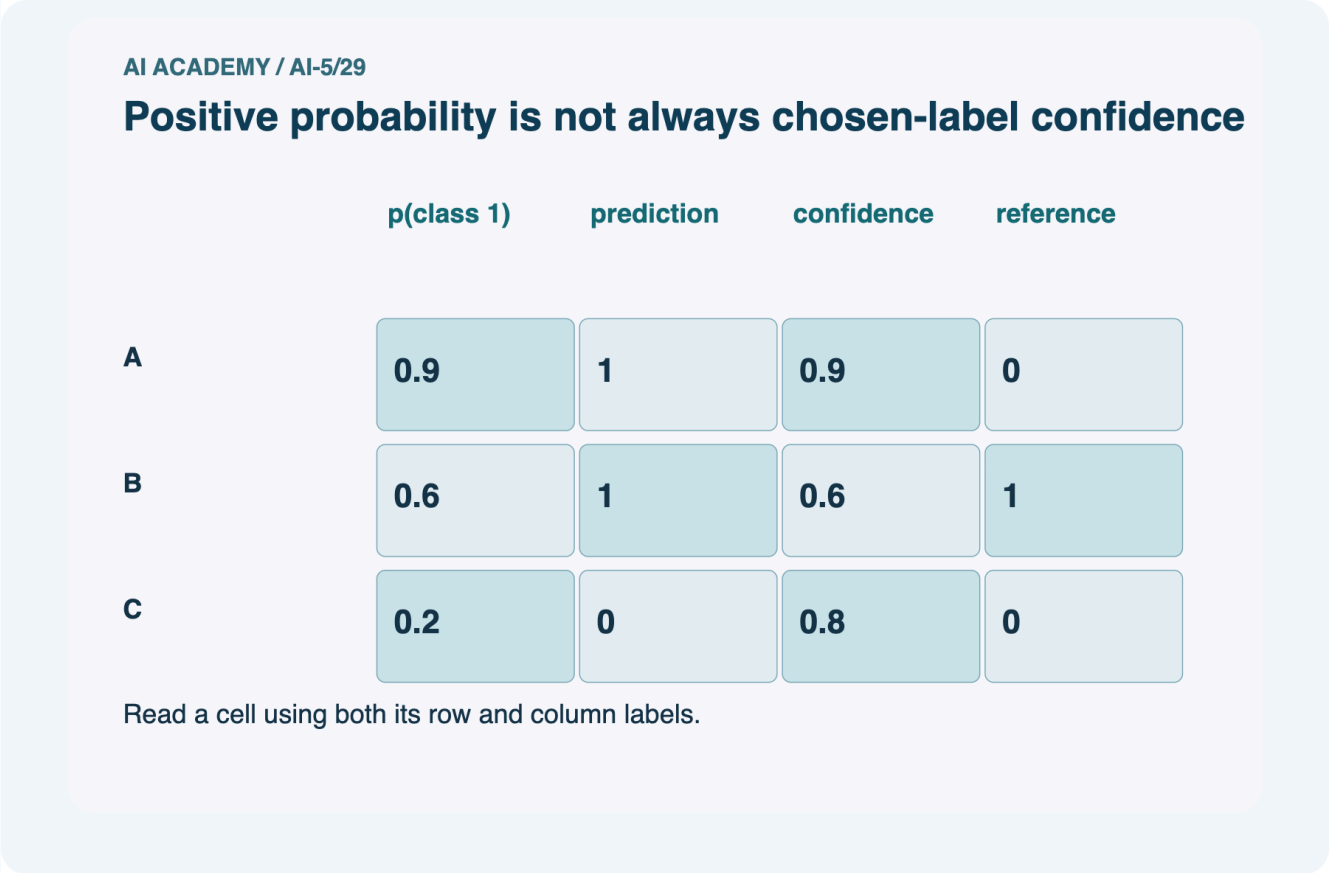
Follow the labelled picture with the matching explanation. Open the vector diagram

Predictions: A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct. → Selected-label confidence: Confidences are 0.9, 0.6, 0.8.

C's confidence is 0.8 despite p(class 1)=0.2. → Selective review: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

03. Positive probability is not always chosen-label confidence



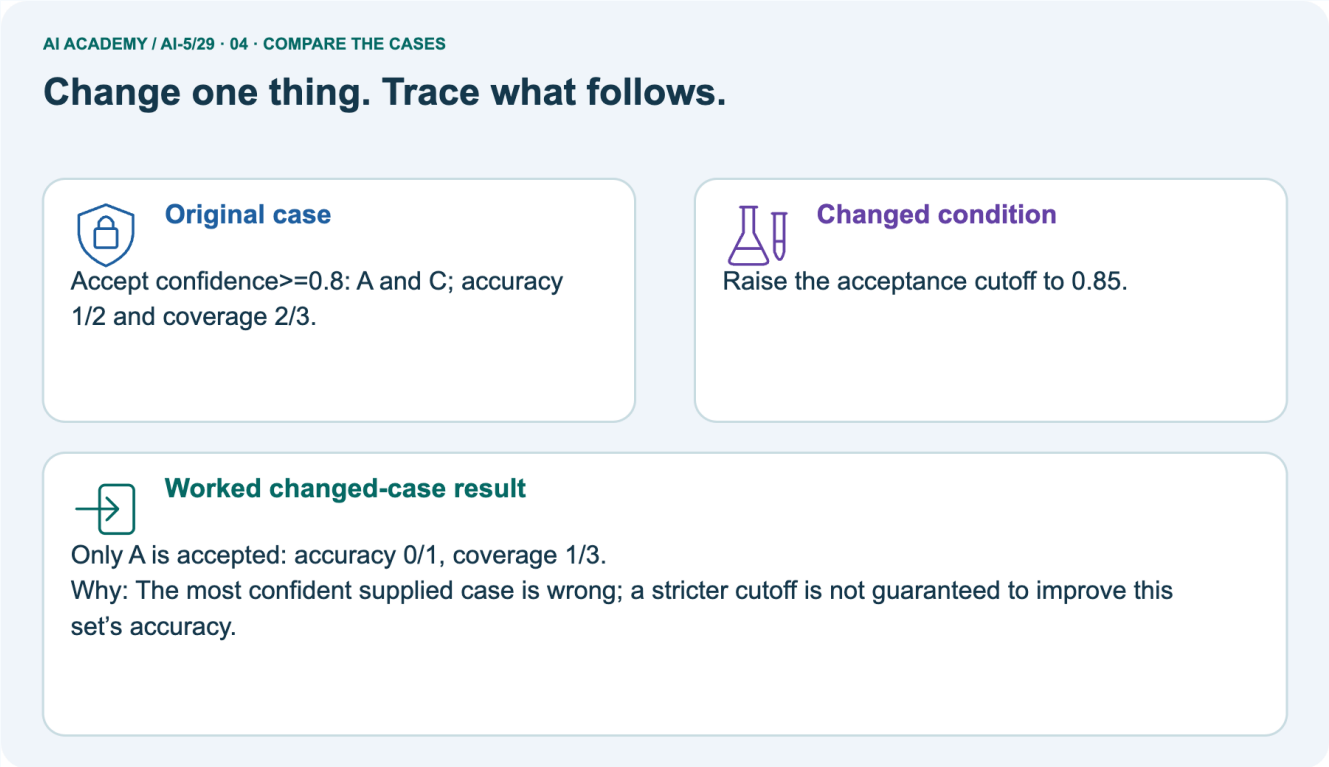
Follow the labelled picture with the matching explanation. Open the vector diagram

Positive probability is not always chosen-label confidence. A: p(class 1) = 0.9, **prediction** = 1, confidence = 0.9, reference = 0. B: p(class 1) = 0.6, **prediction** = 1, confidence = 0.6, reference = 1.

C: p(class 1) = 0.2, **prediction** = 0, confidence = 0.8, reference = 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



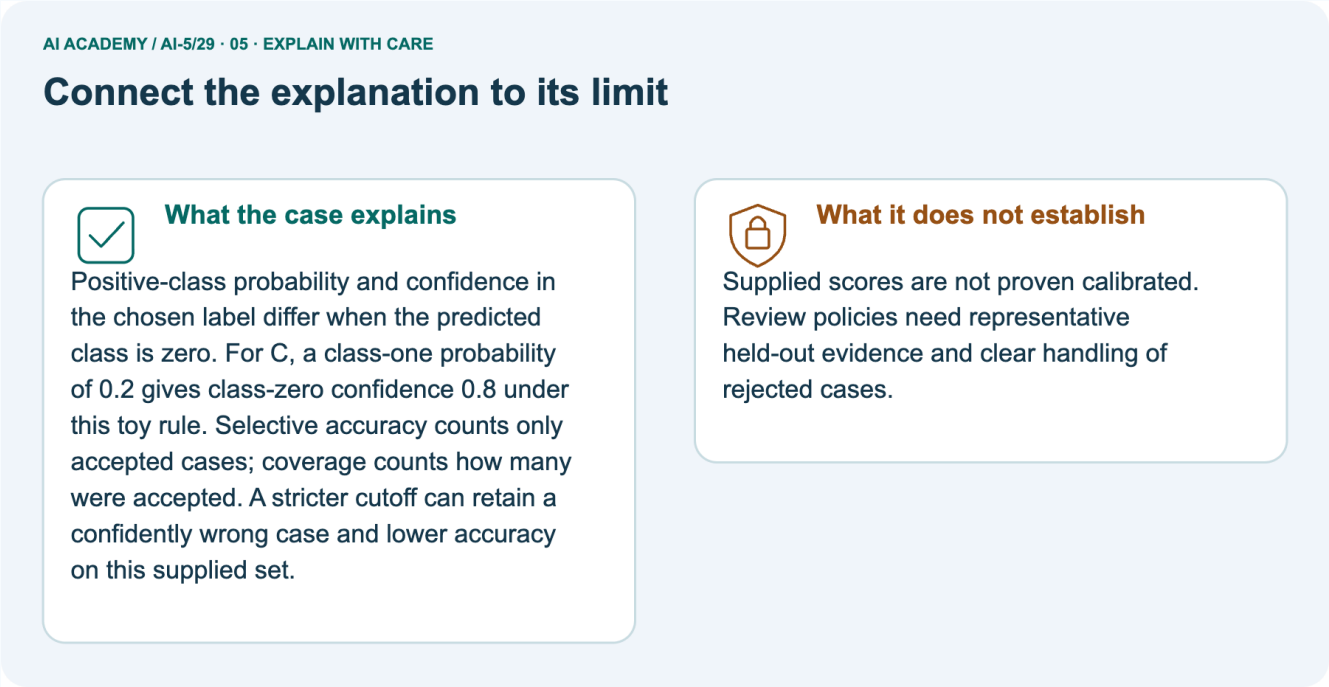
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3. Changed condition: Raise the acceptance cutoff to 0.85. Expected result: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

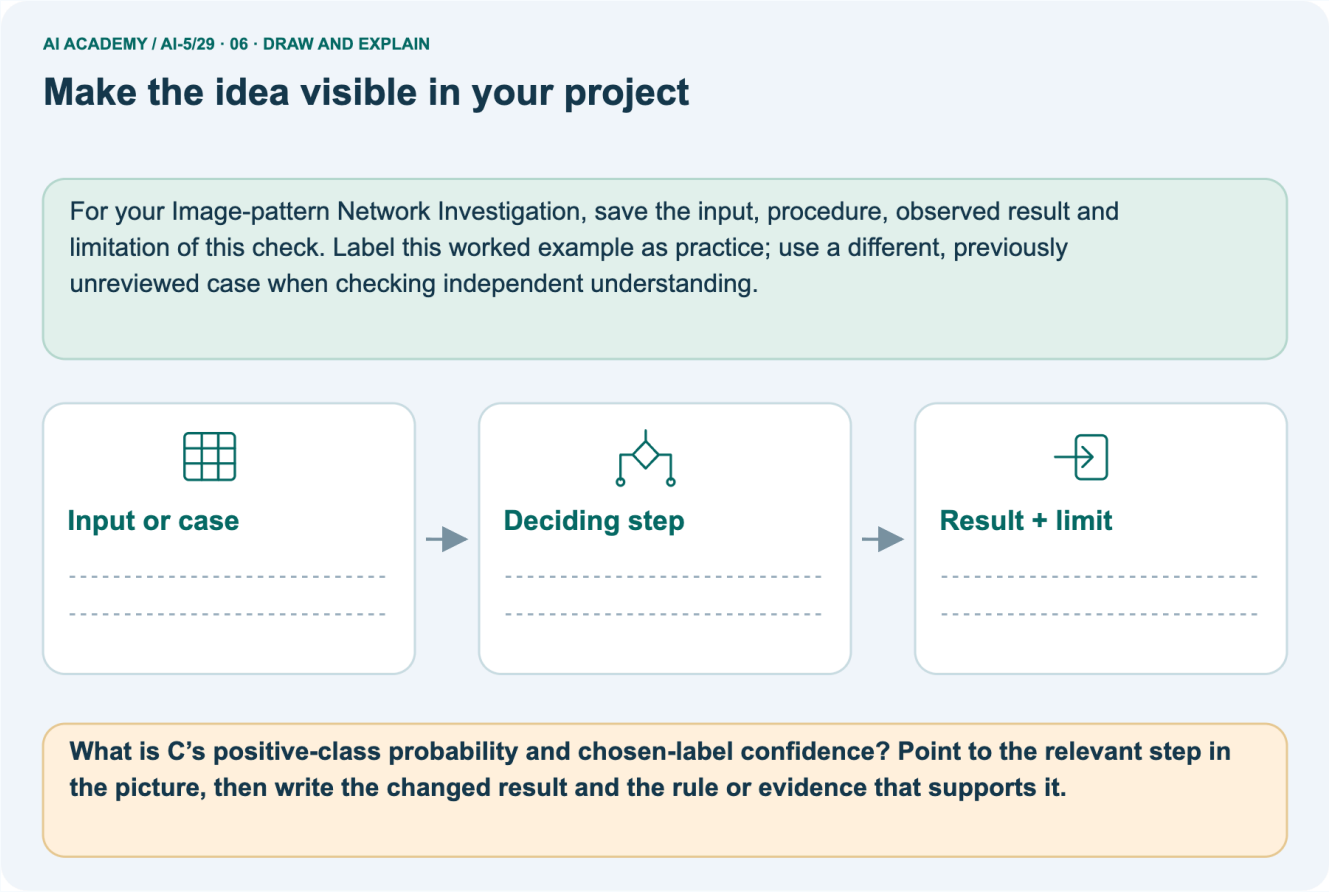
Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Limit: Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Uncertainty and failure analysis: the case

02. Follow the deciding steps

03. Positive probability is not always chosen-label confidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.